

REPORT

ASSIGNMENT 3

GROUP 14

ANUSHKA GUPTA

220101014

KASHISH AGGARWAL

220101050

MAHEK CHOUDHARY

220101062

MAHUA SINGH

220101063

EXCESS - 3 CODE

The excess-3 code (or XS3) is a non-weighted code used to express code used to express decimal numbers. It is a self-complementary binary coded decimal (**BCD**) code and numerical system which has biased representation. It is particularly significant for arithmetic operations as it overcomes shortcomings encountered while using 8421 **BCD code to add two decimal digits** whose sum exceeds 9. Excess-3 arithmetic uses a different algorithm than normal non-biased BCD or binary **positional number system**.

Decimal Digit	BCD Code	Excess-3 Code
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
7	0111	1010
8	1000	1011
9	1001	1100

CYCLIC CODE

A subtype of linear block codes is binary cyclic codes. They have a lot of important features. Errors caused by bursts of noise that influence multiple consecutive bits can be corrected using cyclic codes. Cyclic codes include **Hamming**, **BCH**, and **Golay** codes, all of which are excellent block codes. If every cyclic shift of the code vector creates another code vector, the linear block code is said to be cyclic.

The characteristics of a cyclic code are as follows.

(i) **Linearity Property:** If the modulo-2 sum of any two codewords produces another acceptable codeword, the code is said to be linear.

(ii) **Cyclic Property:** If every cyclic shift of a codeword yields another acceptable code word, the code is said to be cyclic.

Decimal	Cyclic Code			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	0	1	0
4	0	1	1	0
5	1	1	1	0
6	1	0	1	0
7	1	0	0	0
8	1	1	0	0
9	0	1	0	0

Converting EXCESS-3 to CYCLIC

We converted EXCESS-3 to Cyclic code by drawing K-Maps for bit and deriving the corresponding logical expression for the same. Afterwards, we used AND and OR gates to implement the logic. The K-Maps for each bit and their corresponding logical expressions are as follows ->

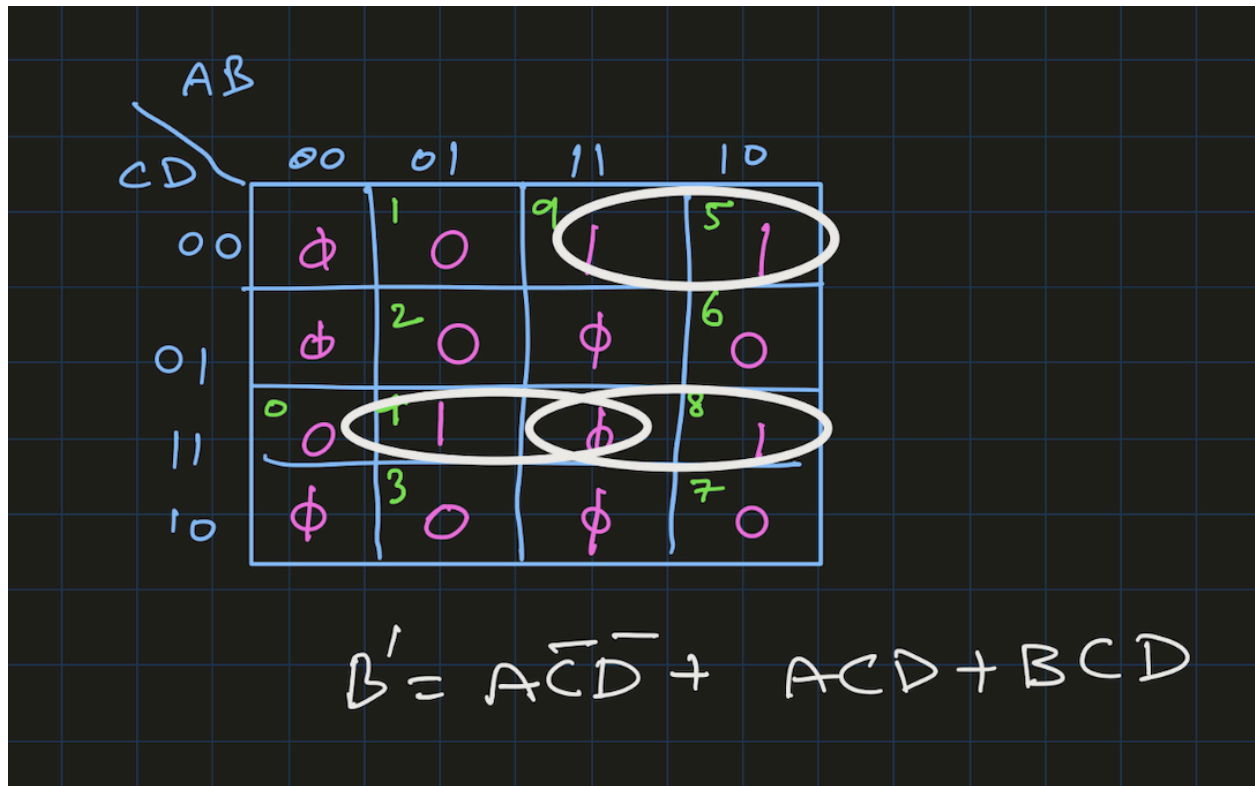
Decimal digit	Excess-3				Cyclic			
0	0	0	1	1	0	0	0	0
1	0	1	0	0	0	0	0	1
2	0	1	0	1	0	0	1	1
3	0	1	1	0	0	0	1	0
4	0	1	1	1	0	1	1	0
5	1	0	0	0	1	1	1	0
6	1	0	0	1	1	0	1	0
7	1	0	1	0	1	0	0	0
8	1	0	1	1	1	1	0	0
9	1	1	0	0	0	1	0	0

AB

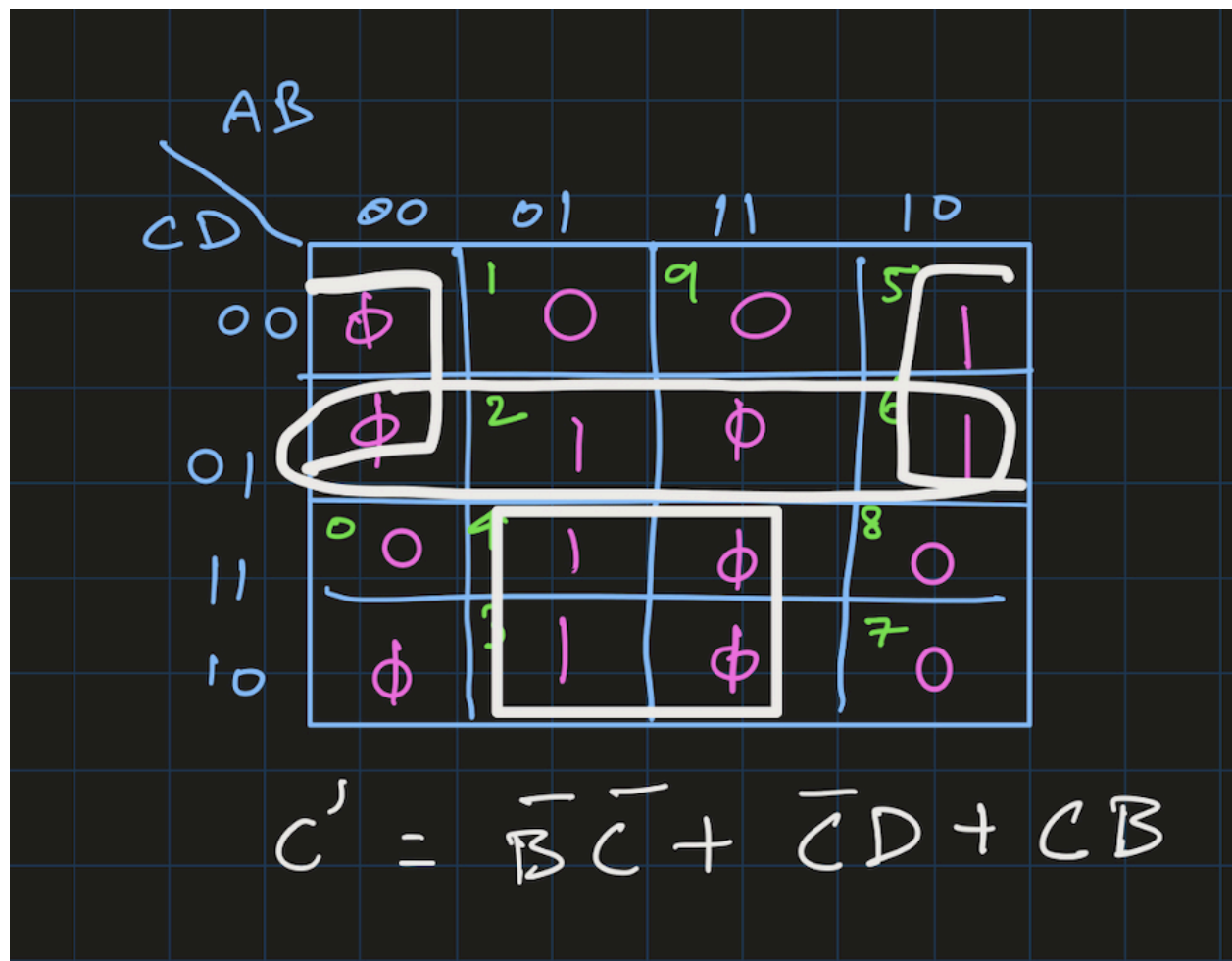
CD \ AB	00	01	11	10
00	ϕ	1 0	9 0	5 1
01	ϕ	2 0	ϕ	6 1
11	0 0	4 0	ϕ	8 1
10	ϕ	3 0	ϕ	7 1

$$A' = A\overline{B}$$

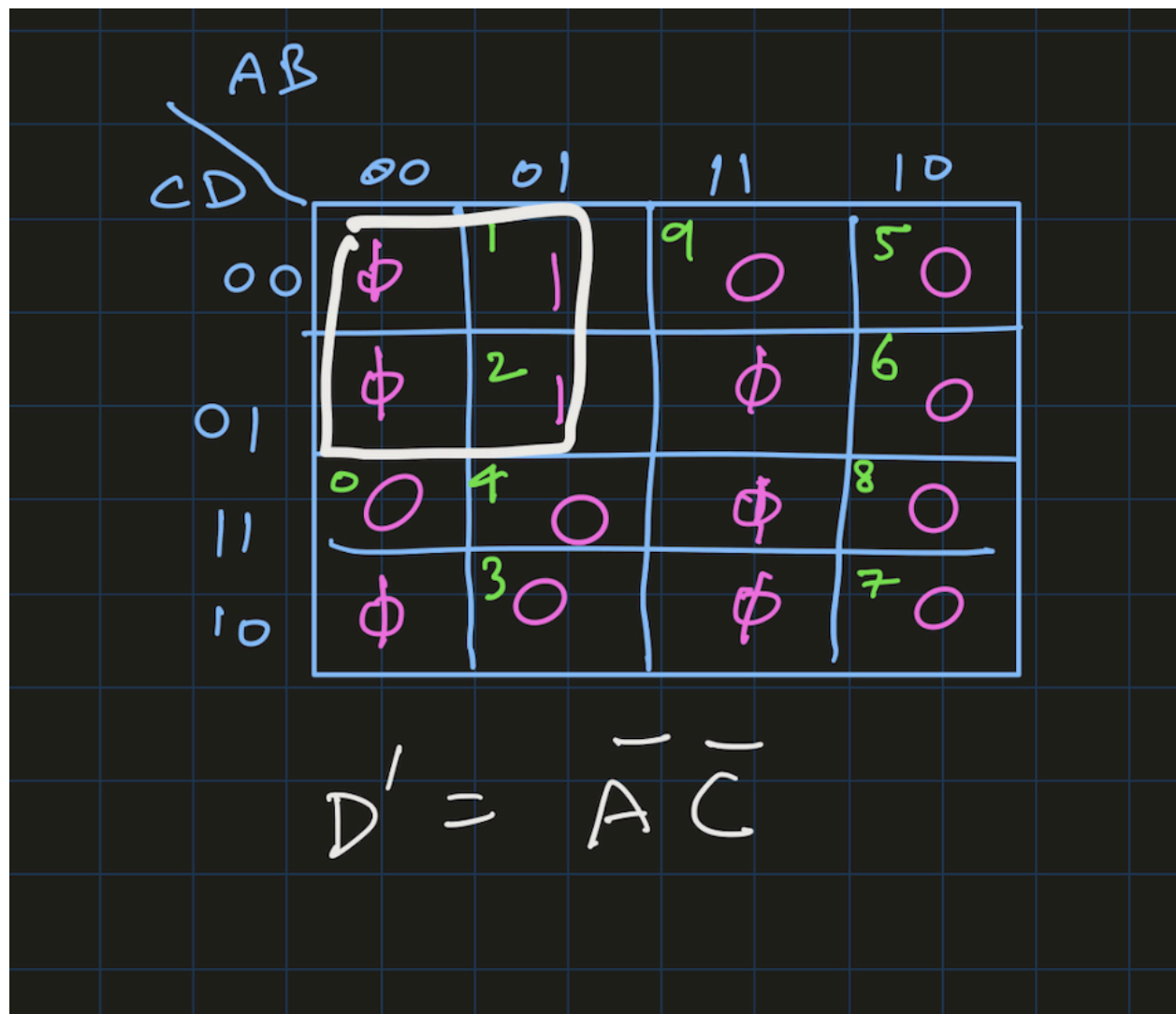
$$A^1 = A\overline{B}$$



$$B^1 = A\bar{C}\bar{D} + ACD + BCD$$



$$C^1 = \overline{B}\overline{C} + \overline{C}D + CB$$



$$D^1 = \overline{A} \overline{C}$$

ICs Used In Implementation

HD74LS08P

The HD74LS08P is like a tiny electronic brain that helps computers and other devices make decisions. It has four small sections called AND gates inside it. These gates are like switches that only turn on if both of their inputs are turned on.

Here's what you need to know:

AND Gates: These are like doors with two switches that both need to be pressed for the door to open.

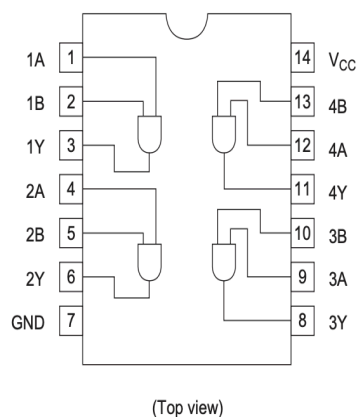
Low Power: It doesn't use a lot of electricity, which is good for saving energy.

Operating Voltage: It needs around 5 volts of electricity to work properly.

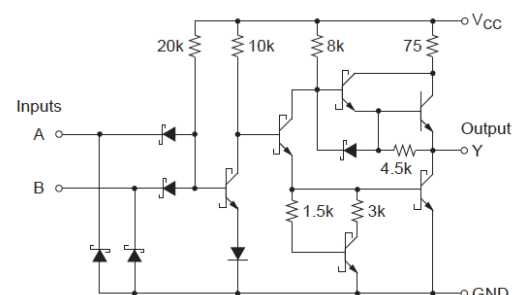
Pin Configuration: It has specific pins for connecting to power, ground, and for sending signals in and out.

What It's Used For: It's used in all sorts of electronic gadgets and devices where decisions need to be made based on multiple conditions.

Pin Arrangement



Circuit Schematic



HD74LS32P

The HD74LS32P is like a tiny electronic part that helps devices make decisions. It has four little sections called OR gates inside it. These gates act like switches: if either switch (or both) is turned on, the output is also turned on.

Here's what you need to know:

OR Gates: These are like doors with two switches, and if at least one switch is on, the door opens.

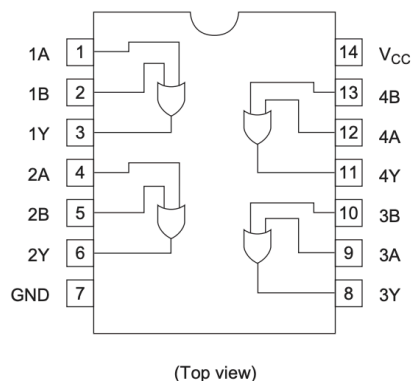
Low Power: It doesn't use a lot of electricity, which helps save energy.

Operating Voltage: It typically needs about 5 volts of electricity to work well.

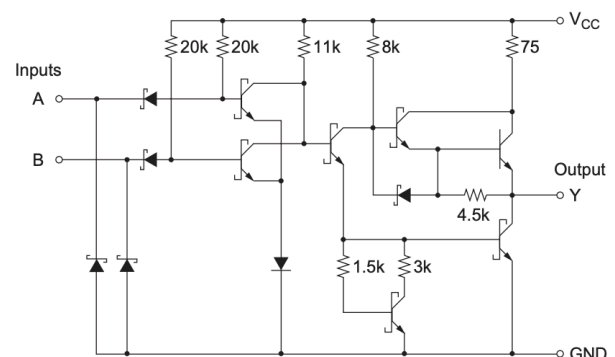
Pin Configuration: It has specific pins for connecting to power, ground, and for sending signals in and out.

Usage: It's used in various electronic devices where decisions need to be made based on different conditions, like whether a button is pressed or a sensor detects something.

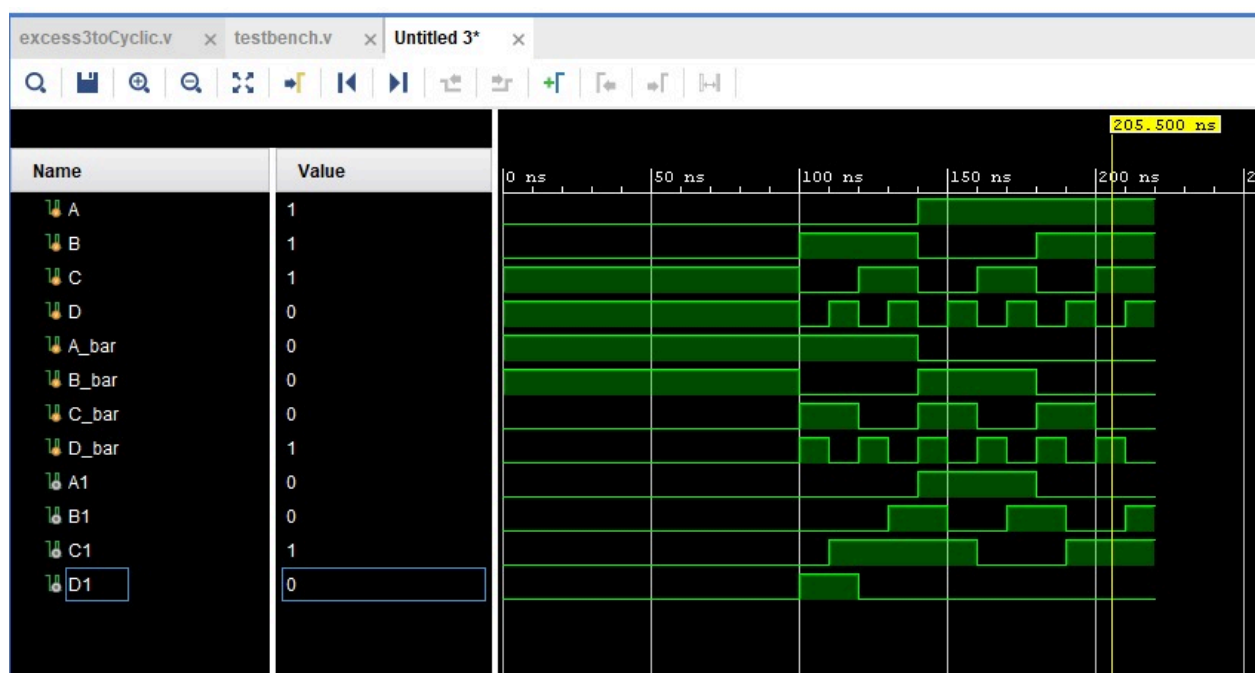
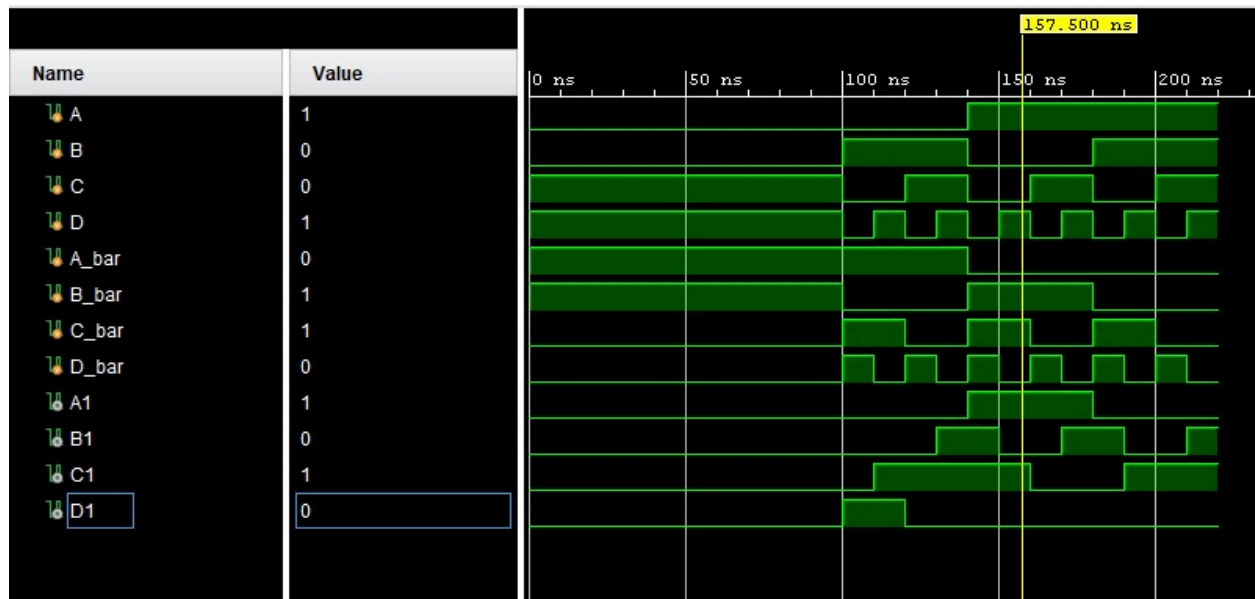
Pin Arrangement



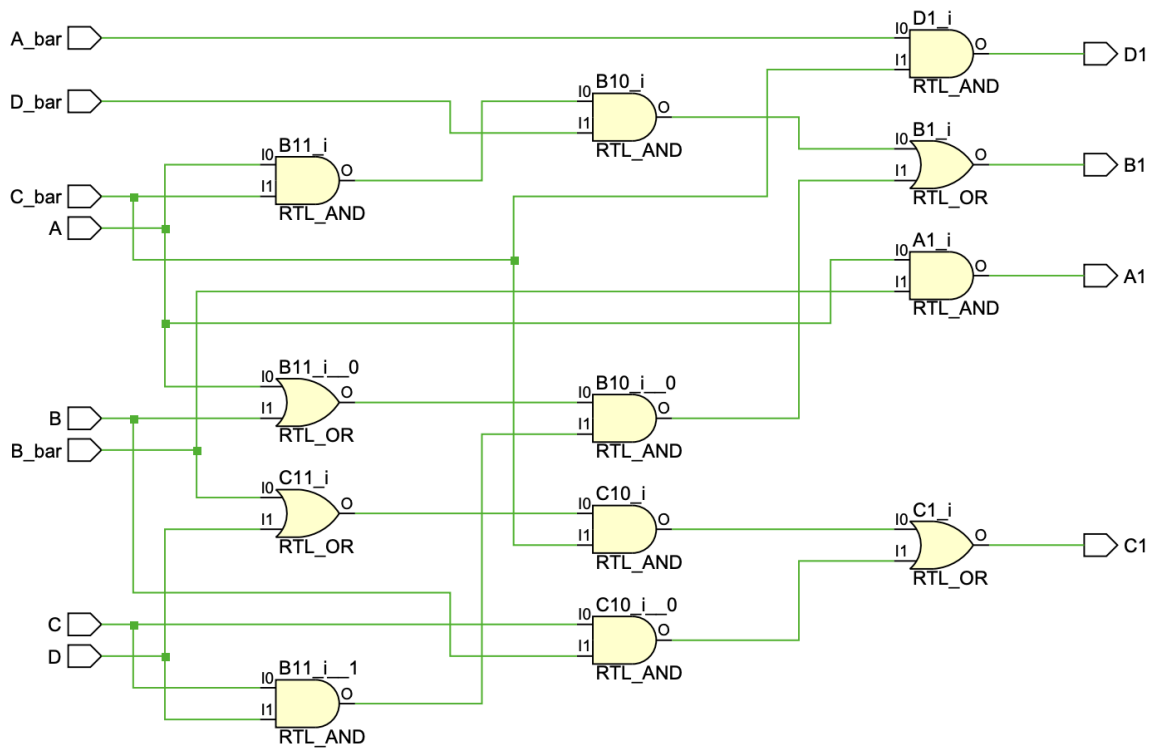
Circuit Schematic



SIMULATION



SCHEMATIC



The image shows a MicroLab-1U digital logic trainer kit. The breadboard contains a 74181 ALU and a 74180 MUX IC. The ALU is configured to perform the operation $A + B$. The MUX is configured to perform the operation $A + B$. The breadboard is populated with logic components and labeled with expressions like $A + B$, $A + B$, and $A + B$. The kit includes a Function Generator, Clock Generator, Counter, Voltmeter, and BCD Display. The board is branded 'MICROLAB-1U TIRUPATI ELECTRONICS'.

